Saveetha Engineering College, Chennai
B.E Computer Science (IoT) | Batch 2022 – 2026

Smart Wastewater pH Monitoring & Alert System

Project Report

Technology Stack

Platform:	Blynk IoT Cloud
Microcontroller:	ESP32 Dev Board
Sensor:	Analog pH Sensor
Language:	Embedded C / Arduino IDE 2.3.7

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1. Abstract

The Smart Wastewater pH Monitoring and Alert System is designed to continuously monitor the pH level of wastewater in real time and provide instant alerts when abnormal conditions are detected, ensuring environmental safety and efficient water management.

2. About the Project

This IoT-based project monitors the acidity and alkalinity of wastewater using real-time sensors and automated alert mechanisms. Wastewater from industries and residential areas can significantly impact the environment if pH levels exceed permissible limits.

Traditional monitoring relies on manual sampling and laboratory testing — time-consuming and prone to delays. This project integrates pH sensors, an ESP32 microcontroller, and Blynk cloud monitoring to provide continuous observation and instant alerts, enabling timely corrective action.

3. Key Features

- Real-time pH monitoring every 2 seconds via ESP32 ADC
- Automated Blynk push alert when pH goes below 6.5 or above 8.5
- IoT-based cloud integration — data visible on Blynk mobile dashboard
- Alert flag prevents repeated notifications until pH returns to safe range
- Serial Monitor output for real-time debugging and verification
- Low power, low cost — runs on USB 5V power supply
- Scalable design suitable for residential, industrial, and agricultural use

4. System Requirements

Hardware Components

Component	Specification
Microcontroller	ESP32 Dev Board (Wi-Fi + ADC built-in)
pH Sensor	Analog pH Sensor Module
Breadboard	Standard 830-point breadboard
Jumper Wires	Male-to-male connection wires
Power Supply	USB 5V via laptop or adapter

Software & Tools

Tool / Platform	Purpose
Arduino IDE 2.3.7	Writing and uploading embedded C code to ESP32
Blynk IoT Platform	Mobile dashboard and push notification alerts
Wi-Fi (ESP32 built-in)	Transmitting sensor data to Blynk cloud
Windows 10 / Ubuntu	Development operating system

5. Hardware Reference — ESP32 Pinout

The pH sensor analog output wire is connected to GPIO 36 (ADC1_0) on the ESP32 Dev Board. GPIO 36 is an input-only ADC pin — ideal for analog sensor readings as it does not interfere with other peripheral operations.

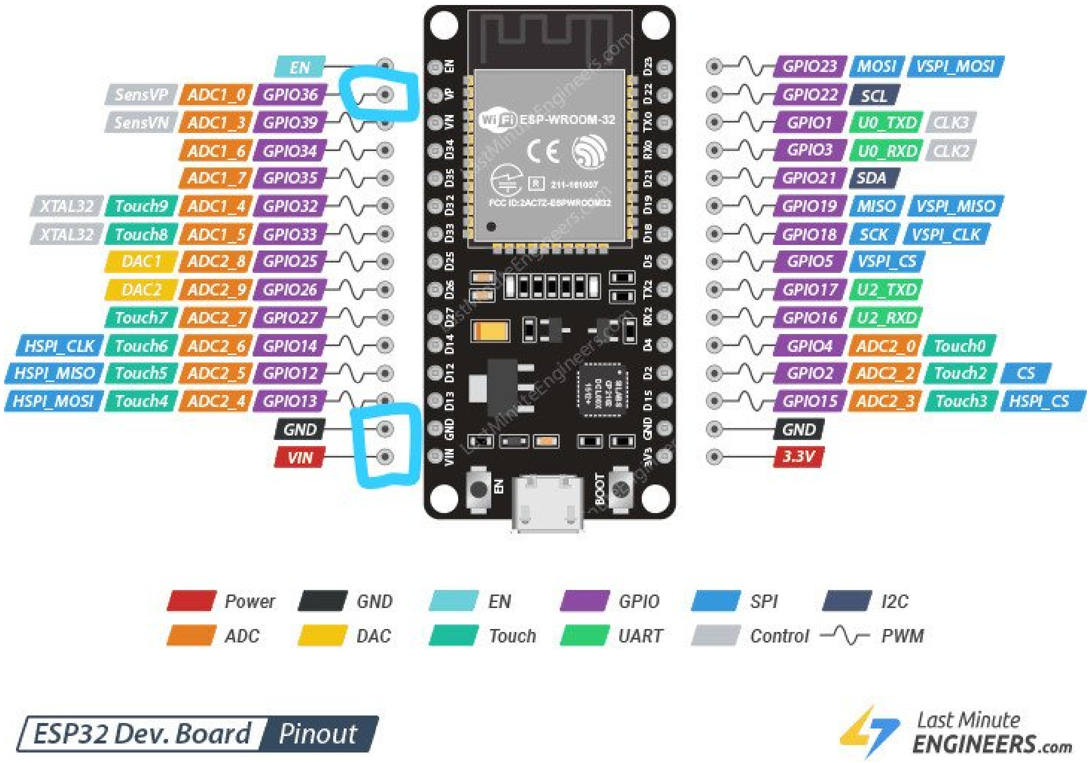

Figure 1 — ESP32 Dev Board Pinout. GPIO 36 (ADC1_0) is used for the pH Sensor analog input.

6. Hardware Setup & Output

The pH sensor probe is submerged in the wastewater sample (blue bowl). The ESP32 Dev Board (red board, left) reads the analog voltage from the pH sensor module (green board, right) via the breadboard. When an unsafe pH is detected, Blynk sends an instant push notification to the owner's mobile phone.

Figure 2 — Physical Hardware Setup (ESP32 + pH Sensor + Bowl)

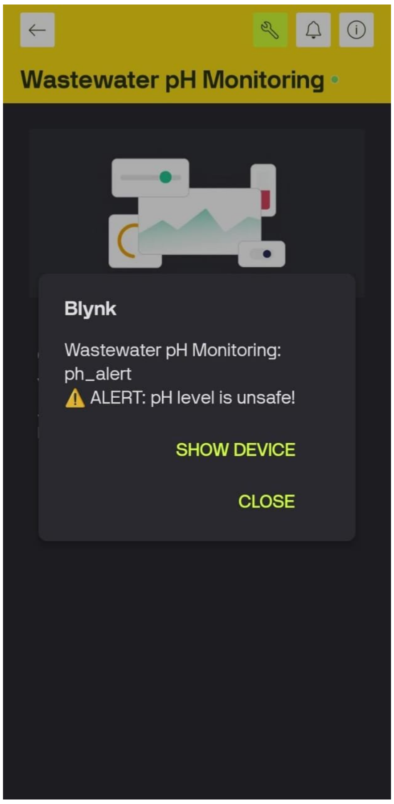

Figure 3 — Blynk Mobile Alert: pH Level Unsafe

7. Source Code

The complete Arduino sketch is shown below. Wi-Fi credentials and Blynk auth token are replaced with placeholders for security. The pH value is read every 2 seconds, sent to Blynk virtual pin V0, and a *ph_alert* event fires when the value is outside 6.5 – 8.5.

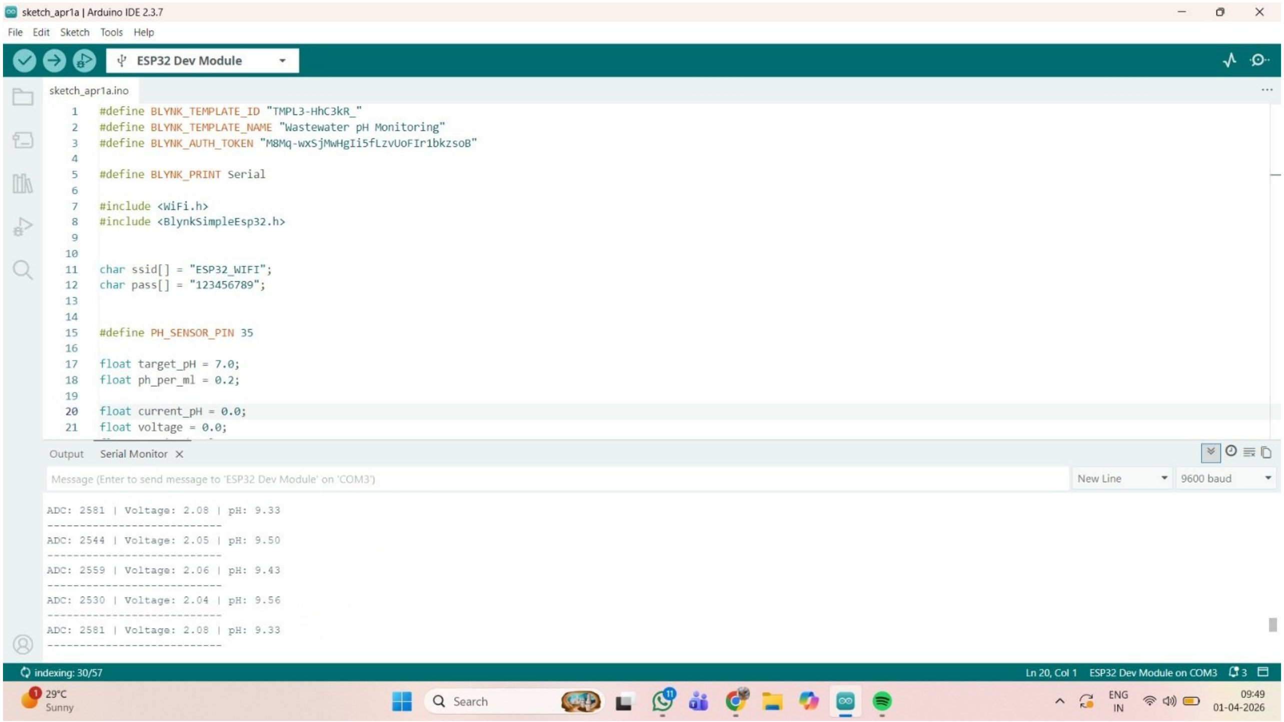

Figure 4 — Arduino IDE 2.3.7 sketch with Serial Monitor output showing live ADC, Voltage and pH readings.

Key Code Logic

```
#define PH_PIN 36           // GPIO 36 — ADC1 input-only pin
float PH_LOW  = 6.5;        // Safe range lower bound
float PH_HIGH = 8.5;        // Safe range upper bound
bool  alertSent = false;    // Prevents repeated alerts

void readPH() {
  int   adc     = analogRead(PH_PIN);
  float voltage = adc * (3.3 / 4095.0);          // 12-bit ADC
  float pHValue = 7 + (2.5 - voltage) * 3.5;     // pH formula

  Blynk.virtualWrite(V0, pHValue);                // Send to dashboard

  if ((pHValue < PH_LOW || pHValue > PH_HIGH) && !alertSent) {
    Blynk.logEvent("ph_alert", "Alert: pH Level Out of Safe Range!");
    alertSent = true;     // Block duplicate alerts
  }
  if (pHValue >= PH_LOW && pHValue <= PH_HIGH) alertSent = false;
}
```

8. Serial Monitor Output

The Serial Monitor in Arduino IDE shows the raw ADC value, converted voltage, and calculated pH value every 2 seconds. Sample readings from the actual test run are shown below:

ADC Value	Voltage (V)	pH Value	Status
2581	2.08 V	9.33	UNSAFE — Alert Sent
2544	2.05 V	9.50	UNSAFE
2559	2.06 V	9.43	UNSAFE
2530	2.04 V	9.56	UNSAFE
1820	1.47 V	7.23	SAFE

9. Conclusion

The Smart Wastewater pH Monitoring and Alert System successfully demonstrates a low-cost, real-time IoT solution for monitoring water quality. The system accurately reads pH levels, transmits data to the Blynk cloud, and alerts users instantly when values are unsafe.

This project has practical applications in residential areas, small industries, and agricultural settings. Future enhancements can include automatic chemical dosing control, multi-node sensor networks, and a web-based analytics dashboard for historical pH trend analysis.